

# Yao Hu

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## Bio

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Yao Hu is currently a Research Assistant Professor with the [Department of Data Science and Artificial Intelligence \(DSAI\)](#) at [The Hong Kong Polytechnic University](#). Prior to that, he was a Postdoctoral Fellow of DSAI under the supervision of Prof. [Kay Chen Tan](#) and Dr. [Jibin Wu](#), as a member of the [Machine Intelligence and Nature-Inspired Computing \(MIND\) Lab](#). He received the B.S. degree in mining engineering and the M.Sc. degree in control science and engineering from the [China University of Mining and Technology](#) under the supervision of Prof. [Xiaoyan Sun](#), and the Ph.D. degree at the [City University of Hong Kong](#) under the supervision of Prof. [Kay Chen Tan](#) and Prof. [Linqi Song](#).

His current research focus is on the data scarcity problem, privacy-preserving domain adaptation, federated learning, and applied deep learning for medical image analysis.

## Education

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**City University of Hong Kong (CityU)**

**Hong Kong SAR, China**

*Ph.D.in Computer Science*

*Sep 2020 - Oct 2024*

- **Research direction:** Data scarcity, Domain adaptation, Medical image analysis
- **Supervisors:** Prof. Kay Chen Tan and Prof. Linqi Song

**China University of Mining and Technology (CUMT)**

**Xuzhou, China**

*M.Sc. in Control Science and Engineering*

*Sep 2017 - Jun 2020*

- **Research direction:** Time series analysis, Federated learning
- **Supervisor:** Prof. Xiaoyan Sun

**China University of Mining and Technology (CUMT)**

**Xuzhou, China**

*B.S. in Mining Engineering*

*Sep 2013 - Jun 2017*

## Research Outputs

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### Journals

- J1. [Yao Hu](#), Y. Huang, R. Liu, X. Xue, Y. Zhang, J. Wu, Z. Huang, K. C. Tan  
“Enhancing X-ray Image Classification through Heterogeneous Federated Learning with Natural Image-Augmented Models”  
*IEEE Transactions on Pattern Analysis and Machine Intelligence*, Early access, 2026 (impact factor:20.3).
- J2. [Yao Hu](#), Z. Huang, R. Liu, X. Xue, X. Sun, L. Song, K. C. Tan  
“Source Free Semi-Supervised Transfer Learning for Diagnosis of Mental Disorders on fMRI Scans”  
*IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 11, pp. 13778–13795, 2023 (impact factor:23.6).
- J3. Y. Huang, [Yao Hu](#) (co-first author), Y. Li, X. Cao, X. Li, K. C. Tan, Z. You, Z. Huang  
“scBIT: Integrating Single-cell Transcriptomic Data into fMRI-based Prediction for Alzheimer’s Disease Diagnosis”  
*IEEE Transactions on Medical Imaging*, vol. 45, no. 7, pp. 3424-3437, 2026 (impact factor:12.4).
- J4. [Yao Hu](#), X. Sun, Y. Tian, L. Song, K. C. Tan  
“Communication Efficient Federated Learning With Heterogeneous Structured Client Models ”  
*IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 7, no. 3, pp. 753-767, 2022 (impact factor:5.3).
- J5. Z. Huang, [Yao Hu](#) (corresponding author), R. Liu, X. Xue, L. Song, Z. Zhu, K. C. Tan  
“Federated Multi-task Learning for Joint Diagnosis of Multiple Mental Disorders on MRI Scans ”  
*IEEE Transactions on Biomedical Engineering*, vol. 70, no. 4, pp. 1137-1149, 2023 (impact factor:4.54).
- J6. [Yao Hu](#), X. Sun, X. Nie, Y. Li  
“An enhanced LSTM for trend following of time series ”  
*IEEE Access*, vol. 7, pp. 34020-34030, 2019 (impact factor:4.098).
- J7. X. Xue, [Yao Hu](#), L. Feng, K. Zhang, L. Song, K. C. Tan  
“Surrogate-Assisted Search with Competitive Knowledge Transfer for Expensive Optimization ”  
*IEEE Transactions on Evolutionary Computation*, In press, 2024 (impact factor:11.7).
- J8. Y. Zhang, [Yao Hu](#), C. Cai, Y. Huang, Z.-A., K. C. Tan  
“Anti-Confounding Hashing: Enhancing Radiological Image Retrieval via Debiased Weighting and Counterfactual Reasoning ”  
*IEEE Transactions on Neural Networks and Learning Systems*, vol. 36, no. 8, pp. 15055-15069, 2025 (impact factor:10.4).

- J9. R. Liu, Yao Hu, J. Wu, K.-C. Wong, Z. Huang, Y. Huang, K. C. Tan  
“Dynamic Graph Representation Learning for Spatio-Temporal Neuroimaging Analysis ”  
*IEEE Transactions on Cybernetics*, vol. 55, no. 3, pp. 1121-1134, 2025 (impact factor:9.4).
- J10. R. Liu, Z. Huang, Yao Hu, Z. Zhu, K.-C. Wong, K. C. Tan  
“Spatial-Temporal Co-attention Learning for Diagnosis of Mental Disorders from Resting-State fMRI data ”  
*IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 8, pp. 10591-10605, 2024 (impact factor:10.4).
- J11. Y. Zhang, Y. Huang, Y. Hu, R. Liu, J. Wu, Z. Huang., K. C. Tan  
“CausalMixNet: A mixed-attention framework for causal intervention in robust medical image diagnosis ”  
*Medical Image Analysis*, 103581, 2025 (impact factor:10.7).
- J12. R. Liu, Z. Huang, Yao Hu, Z. Zhu, K.-C. Wong, K. C. Tan  
“Attention-like Multimodality Fusion with Data Augmentation for Diagnosis of Mental Disorders using MRI ”  
*IEEE Transactions on Neural Networks and Learning Systems*, vol. 36, no. 6, pp. 7627-7641, 2024 (impact factor:14.26).
- J13. X. Xue, C. Yang, Yao Hu, K. Zhang, Y. M. Cheung, L. Song, K. C. Tan  
“Evolutionary Sequential Transfer Optimization for Objective-Heterogeneous Problems ”  
*IEEE Transactions on Evolutionary Computation*, vol. 26, no. 6, pp. 1424–1438, 2021 (impact factor:16.497).
- J14. R. Liu, Z. Huang, Yao Hu, L. Huang, K.-C. Wong, K. C. Tan  
“Spatio-Temporal Hybrid Attentive Graph Network for Diagnosis of Mental Disorders on fMRI Time-Series Data ”  
*IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 8, no. 6, 2024 (impact factor:5.3).
- J15. H. He, Z. Tang, G. Chen, F. Xu, Yao Hu, Y. Feng, J. Wu, Y. Huang, Z. Huang, et al.  
“scKAN: interpretable single-cell analysis for cell-type-specific gene discovery and drug repurposing via Kolmogorov-Arnold networks ”  
*Genome Biology*, vol. 26, no. 1, 300, 2025 (impact factor:12.1).

## Conferences

- C1. Q. Lin, X. Sun, Yao Hu (co-corresponding author), W. Fang  
“EvoCSFL: Surrogate-Assisted Evolutionary Client Selection for Efficient and Robust Federated Learning ”  
*2026 International Joint Conference on Artificial Intelligence (IJCAI)*, Bremen, Germany, August 2026.
- C2. C. Cao, X. Xue, L. Feng, Yao Hu (corresponding author), K. C. Tan  
“On the Generalization Gain of Transfer Learning ”  
*2026 International Joint Conference on Neural Networks (IJCNN)*, IEEE, Maastricht, Netherlands, June 2026.
- C3. Yao Hu, R.Liu, J. Wu, Z. Huang, K. C. Tan  
“Federated Learning with Robust Local Training for Noisy MRI-Based Mental Disorder Diagnosis ”  
*2025 International Conference on Machine Intelligence and Nature-Inspired Computing (MIND)*, IEEE, Xiamen, China, Nov. 2025.
- C4. Yao Hu, Z. Huang, R.Liu, J. Zhang, L. Song, K. C. Tan  
“Heterogeneous Structured Federated Learning with Graph Convolutional Aggregation for MRI-Based Mental Disorder Diagnosis ”  
*2024 International Joint Conference on Neural Networks (IJCNN)*, IEEE, Yokohama, Japan, July 2024.
- C5. Yao Hu, Z. Huang, R.Liu, X.Xue, L. Song, K. C. Tan  
“A Dual-stage Pseudo-Labeling Method for the Diagnosis of Mental Disorder on MRI Scans ”  
*2022 International Joint Conference on Neural Networks (IJCNN)*, IEEE, Padua, Italy, July 2022.
- C6. Yao Hu, X. Sun, Y. Chen, Z. Lu  
“Model and Feature Aggregation Based Federated Learning for Multi-Sensor Time Series Trend Following ”  
*International Work-Conference on Artificial Neural Networks*, Springer, Gran Canaria, Spain, July 2019.
- C7. Y. Chen, X. Sun, Yao Hu  
“Federated Learning Assisted Interactive EDA with Dual Probabilistic Models for Personalized Search”  
*International Conference on Swarm Intelligence*, Springer, Chiang Mai, Thailand, July 2019.

## Authorized Patents

- P1. Z. Li, Yao Hu, Z. Wen, Adaptive Text Emotion Recognition Model Training Method, Electronic Device, and Storage Medium, China, ZL 2023 1 0315955.5, Jun 30, 2023.
- P2. X. Sun, Yao Hu, Z. Li, SVD-Based Heterogeneous Structured Federated Learning for Trend Following of Electromagnetic Radiation Intensity Time-Series, China, ZL 2019 1 1125937.0, Jun 20, 2023.
- P3. Z. Li, Yao Hu, Z. Wen, Multimodal Data Fusion Method, Device, Equipment and Mediums for Speech Recognition, China, ZL 2023 1 0029724.8, Jun 2, 2023.
- P4. Z. Li, Yao Hu, Z. Wen, Subtitle Generation Methods, Electronic Devices, and Storage Mediums Based on Federated Learning, China, ZL 2022 1 1192216.3, Mar 24, 2023.
- P5. Z. Li, X. Xie, Yao Hu, Z. Wen, Disease Diagnosis Methods, Devices, Equipment, and Mediums Based on Multi-Modal Data, China, ZL 2023 1 0550630.5, Jul 25, 2023.

P6. X. Sun, L. Liu, Yao Hu, Selection and Trend Following Method of Electromagnetic Radiation Intensity Time-Series, China, ZL 2018 1 0498234.1, Feb 08, 2022.

### ***Manuscripts in Submission***

- M1. Yao Hu, R. Liu, X. Xue, Y. Zhang, X. Sun, J. Wu, Z. Huang, K. C. Tan  
“A Bayesian Robust Federated Learning Framework for MRI-Based Mental Disorder Diagnosis Under Noisy Labels”  
*IEEE Computational Intelligence Magazine* (Major Revision, 1st round).
- M2. Yao Hu, Z. Huang, R. Liu, X. Xue, J. Wu, Y. Liu, L. Song, K. C. Tan  
“Bridging the Gap: Natural Images for X-Ray Diagnosis on Scarce Samples ”  
*IEEE Transactions on Neural Networks and Learning Systems* (Under Review, 2nd round).
- M3. F. Xu, Z. Huang, H. He, Y. Song, W. Liu, D. Zhang, Yao Hu (co-corresponding author), K. C. Tan  
“AbLWR: A Context-Aware Listwise Ranking Framework for Antibody-Antigen Binding Affinity Prediction via Positive-Unlabeled Learning ”  
*Bioinformatics* (Under Review).
- M4. Y. Zhang, Z. Huang, X. Wu, S. Gao, R. Liu, Z. Chen, J. Wu, Yao Hu (co-corresponding author), K. C. Tan  
“Advancements in Multimodal Foundation Models for Healthcare: An In-Depth Review and Future Outlook ”  
*IEEE Transactions on Neural Networks and Learning Systems* (Under Review).

### **Honors and Awards**

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|----------------------------------------------------|-----------|
| 1. Research Tuition Scholarships of CityU          | 2023-2024 |
| 2. Outstanding Academic Performance Award of CityU | 2022-2023 |
| 3. PhD Scholarship of CityU                        | 2020-2024 |
| 4. Outstanding Graduates of CUMT                   | 2020      |
| 5. Outstanding Master's Thesis of CUMT             | 2020      |
| 6. National Scholarship for Graduate Students      | 2019      |
| 7. Jiangsu Xinjie Enterprise Scholarship           | 2019      |

### **Professional Activities**

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#### ***Conference Community***

Programming Committee Member of the *2026 BIBM Conference*; Organizing Committee Chairs of the *2026 ISPCT Conference*; Technical Program Chairs of the *2026 BIC Conference*; Local Organizing Committee Member of the *2026 ISCAIT Conference*.

#### ***Journal Reviewer***

*IEEE Transactions on Neural Networks and Learning Systems* (TNNLS), *IEEE Transactions on Evolutionary Computation* (TEVC), *IEEE Transactions on Medical Imaging* (TMI), *IEEE Transactions on Industrial Informatics* (TII), *Journal of Biomedical and Health Informatics* (JBHI), *IEEE Transactions on Cognitive and Developmental Systems* (TCDS), *IEEE Transactions on Emerging Topics in Computational Intelligence* (TETCI), *Neural Computing and Applications* (NCAA)

#### ***Conference Reviewer***

*International Conference on Medical Image Computing and Computer Assisted Intervention* (MICCAI), *International Joint Conference on Neural Networks* (IJCNN), *IEEE Conference on Artificial Intelligence* (CAI), *IEEE Big Data*

### **Teaching Experience**

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- Teaching assistant for IT Professionals and Society (CS2066)
- Teaching assistant for Artificial Intelligence – Past, Present, and Future (GE2340)
- Teaching assistant for Internet Security and e-Commerce Protocols (CS4286)
- Teaching assistant for Data Engineering (CS5481)
- Teaching assistant for Machine Learning: Principles and Practice (CS5487)

### **Academic Report**

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- AI Lecture 33 -Jiangnan University (Invited by Prof. Xiaoyan Sun)
- Report at Dongguan University of Technology (Invited by Prof. Hanxu Hou)